

Domain-Oriented Teaching Strategies for Integrating Generative AI into Business Analytics Education

First A. Author, Second B. Author, Jr., and Third C. Author, *Member, IEEE*

Abstract— Generative AI (GenAI) is rapidly entering Analytics education, yet existing applications remain fragmented across tasks and disciplines. This study develops literature-grounded teaching strategies for systematic GenAI integration across the seven domains of the INFORMS Analytics Framework. A corpus of 245 Scopus-indexed publications (2022–2026) was re-analyzed using a domain-oriented, multi-label classification procedure, identifying 91 studies with relevant educational contributions. Evidence was concentrated in Data and Analytics/Model Development, while Analytics Problem Framing, Deployment, and Analytics Solution Lifecycle Management were comparatively underrepresented. For each domain, literature-derived practices were synthesized and complementary teaching strategies were proposed. The resulting strategies specify how GenAI can support the corresponding Analytics competencies while preserving learner responsibility, verification, human oversight, and appropriate evidence of learning. Overall, the strategies position GenAI as structured human-AI support rather than task automation, emphasizing learner agency, analytical judgment, verification, human oversight, and responsible AI use.

Index Terms— Analytics Education, Generative Artificial Intelligence, Human-AI Learning, INFORMS Analytics Framework, Teaching Strategies.

I. INTRODUCTION

BUSINESS Analytics (BA) is a multidisciplinary domain that combines analytical methods, statistical techniques, computational tools, and business knowledge to support data-driven decision-making [1], [2], [3]. Its education integrates competencies in statistics, programming, databases, visualization, machine learning, and decision-making, while increasingly incorporating artificial intelligence technologies [2], [4]. Accordingly, educational evidence from these constituent analytical areas is considered relevant when it addresses competencies represented within the Analytics

process. The rapid rise of Generative AI (GenAI), particularly large language models (LLMs), retrieval-augmented generation (RAG), and AI copilots, further expands these possibilities by enabling natural-language interaction with data, analytical methods, code, and specialized knowledge [5], [6], [7], [8]. As a result, GenAI can support multiple stages of analytical work, including problem formulation, data analysis, programming, visualization, interpretation, and communication.

Recent studies have applied GenAI to statistical analysis and code generation [9], database querying and verification [10], statistical literacy [11], game-based analytical learning [12], and AI-supported decision-making [13], [14]. These contributions indicate a shift from isolated assistance toward GenAI-supported analytical learning and AI-augmented competence [10]. However, these applications provide limited guidance on how GenAI-supported practices can be translated into systematic teaching strategies across the broader set of competencies that constitute Analytics education. In particular, there is a need to determine how GenAI can support different parts of the Analytics process while preserving verification, human oversight, learner agency, and responsible use [9], [10], [13], [15].

Therefore, this study aims to develop literature-grounded teaching strategies for integrating GenAI into Analytics education by synthesizing existing educational practices across the domains of the INFORMS Analytics Framework, proposing complementary strategies, particularly where current educational evidence is limited. Building on a corpus of 245 publications, the study moves from identifying what GenAI applications exist toward establishing how these technologies can be systematically incorporated into Analytics teaching and learning. Section II presents the literature review; Section III describes the research method; Section IV presents the results of the domain-oriented classification; Section V discusses literature-derived and proposed teaching strategies across the Analytics domains; Section VI synthesizes these strategies and outlines educational implications and research opportunities; Section VII presents the limitations of the study; and Section VIII concludes the study.

F. A. Author is with the National Institute of Standards and Technology, Boulder, CO 80305 USA (corresponding author to provide phone: 303-555-5555; fax: 303-555-5555; e-mail: author@boulder.nist.gov).

S. B. Author, Jr., was with Rice University, Houston, TX 77005 USA. He is now with the Department of Physics, Colorado State University, Fort Collins, CO 80523 USA (e-mail: author@lamar.colostate.edu).

T. C. Author is with the Electrical Engineering Department, University of Colorado, Boulder, CO 80309 USA, on leave from the National Research Institute for Metals, Tsukuba, Japan (e-mail: author@nrim.go.jp).

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II. LITERATURE REVIEW

A. Transformation of Analytics Education and the Emergence of GenAI

Analytics education is being reshaped by the increasing capacity of Generative AI (GenAI) to participate in activities that previously required specialized software, programming syntax, or extensive technical mediation. In statistics, large language models (LLMs) can extend conventional workflows by connecting structured analysis with unstructured text through preprocessing, classification, summarization, and feature extraction, while preserving the importance of foundational statistical reasoning [16]. Related curriculum reforms integrate LLMs with software applications, data analysis, and visualization, positioning GenAI as part of broader competency development rather than as a standalone technology [17]. Prompt-driven approaches such as vibe-modeling further illustrate how natural-language interaction can make analytical model development more iterative and accessible, while creating risks of hidden assumptions and superficial sophistication when critical oversight is absent [18]. GenAI is therefore changing not only how analytical activities are performed, but also the competencies and learning experiences required in Analytics education.

B. GenAI-Supported Analytical Learning and Teaching Practices

Existing research demonstrates multiple ways in which GenAI can support the teaching and learning of Analytics. In statistics and chemometrics, students have used LLMs for visualization, hypothesis testing, interpretation, and R-code generation while independently verifying numerical results and methodological choices using conventional software [9]. In Business Analytics, GenAI has been incorporated into regression, forecasting, and hypothesis-testing assignments in which students compare generated outputs with manual Excel analyses and critique their accuracy, ethical implications, and professional relevance [19]. Other educational applications include conversational support for statistical problem solving [20], Copilot-assisted text mining in Excel [21], and GenAI as a complement to no-code machine-learning environments [22].

These applications increasingly position GenAI as a collaborator, tutor, or critique partner rather than merely an answer generator. Nevertheless, access to GenAI does not automatically improve learning. In data-visualization instruction, students perceived GenAI positively, but measured learning outcomes did not differ significantly from those of students without AI support [23]. Programming studies similarly report task-dependent effects [24]. Conversely, explicitly teaching appropriate GenAI use and embedding it within assessment has been associated with improved application performance compared with unstructured access [25]. Together, these findings emphasize that educational value depends on how GenAI is integrated into learning activities, including scaffolding, verification, assessment, and preservation of learner responsibility.

C. INFORMS Analytics Domains as an Organizing Lens for GenAI Education

Although GenAI applications are often studied within individual subjects or tasks, Analytics education prepares learners to address interconnected activities across an Analytics process. Project-based educational studies illustrate this broader perspective. One data-science study incorporated ChatGPT throughout a modified CRISP-DM process, from idea generation to deployment, and found that students perceived its usefulness differently across stages [26]. A Business Analytics teaching case similarly combined ChatGPT with CRISP-DM to support data understanding, logistic-regression modeling, interpretation, and development of a business solution [27]. Other cases integrate GenAI across forecasting, financial analysis, variance analysis, investment assessment, and professional communication [28], or across relational design, SQL generation, feedback, and database maintenance [29].

These studies suggest that GenAI can contribute to learning in different parts of the Analytics process, but its educational role, appropriate level of assistance, and required learner competencies may vary substantially between them. Accordingly, the present study uses the INFORMS Analytics Framework [30] as an organizing lens for Analytics education, covering Business Problem Framing, Analytics Problem Framing, Data, Methodology Selection, Analytics/Model Development, Deployment, and Analytics Solution Lifecycle Management. Rather than treating the framework as a project-management methodology, these domains provide an Analytics-specific structure for examining where and how GenAI can support the development of competencies needed to conduct Analytics work.

D. Research Gap and Positioning of the Present Study

Despite rapid growth in applications, research on GenAI in Analytics education remains fragmented across analytical subjects, technologies, courses, and pedagogical interventions. Reviews of database education, for example, identify relatively few empirical studies and a predominance of supplementary assistance rather than deeper forms of educational integration [10]. Research on LLM tutoring also shows that accurate responses may undermine learning when complete solutions are provided before students engage in productive problem solving [31]. Similarly, high numerical accuracy in statistical tasks does not necessarily imply conceptual reliability or enduring learning [32]. Thus, existing research provides valuable evidence for particular tasks, but limited guidance for systematically integrating GenAI across the broader set of competencies that constitute Analytics education.

The present study addresses this gap by reexamining GenAI applications in Analytics education through the seven INFORMS domains. It identifies how existing educational practices contribute to learning within each domain, derives transferable teaching strategies from the literature, and proposes complementary strategies where current evidence is limited. These literature-derived and proposed strategies are

subsequently synthesized as a structured set of strategies for GenAI-supported Analytics education, with particular attention to learner responsibility, verification, human oversight, and responsible AI use.

III. MATERIALS AND METHODS

The study follows a structured workflow comprising four stages: (1) adoption of the evidence base, (2) re-analysis of the selected publications using the INFORMS Analytics domains, (3) synthesis and extension of GenAI-supported teaching strategies, and (4) organization of these strategies into a domain-oriented set for GenAI-supported Analytics education. The INFORMS domains are used as an Analytics-specific organizing lens to examine how GenAI can contribute to developing competencies associated with different parts of Analytics work, rather than as a project-management methodology.

A. Study Design and Evidence Base

The evidence base was drawn from a corpus the authors previously developed as part of a broader state-of-the-art review of GenAI in Analytics education [33]. That study used Scopus and an iterative search strategy organized around three conceptual blocks: GenAI, Analytics, and education. The query combined terminology related to GenAI and specific models with Business Analytics, Data Analytics, constituent analytical areas, and educational terms associated with teaching, learning, curriculum, and higher education. The final search was conducted on July 10, 2026. Following the screening procedure described in [34], [35], the unrestricted search retrieved 5,785 records. After removing 174 records according to language and document-type criteria, 5,611 publications remained for eligibility screening. Titles and abstracts were assessed for a substantive relationship among GenAI, Analytics, and education, resulting in 5,366 exclusions and a final corpus of 245 Scopus-indexed publications from 2022-2026.

Whereas the companion manuscript [33] examines the field's conceptual structure and emerging themes through descriptor-level bibliometric and co-word analyses, the present research re-analyzes individual publications in this corpus to determine how their reported educational contributions can inform GenAI-supported Analytics education within the seven INFORMS domains.

B. Domain-Oriented Classification

The 245 publications were classified according to Business Problem Framing, Analytics Problem Framing, Data, Methodology Selection, Analytics/Model Development, Deployment, and Analytics Solution Lifecycle Management. For each domain, the analysis addressed the question: What grounded contribution does this publication provide for using GenAI to teach or learn competencies associated with this Analytics domain?

Classification followed a four-level relevance procedure. An internal scale (absent, resemblance only, clear evidence, and explicit evidence) was separately applied to each domain.

“Absent” indicated no relevant educational contribution to the domain; “resemblance only” indicated a related activity without sufficient evidence that the corresponding Analytics competency was being taught or learned; “clear evidence” indicated a supported educational contribution associated with the domain; and “explicit evidence” indicated that the corresponding competency was directly taught, practiced, assessed, or examined. A contribution was retained for subsequent synthesis only when it reached the clear-evidence or explicit-evidence level by describing a relevant educational use, learner activity, scaffold, observed behavior, benefit, limitation, or risk. Classification was multi-label; consequently, a publication could contribute to none, one, or several domains. Therefore, the complete set of 245 publications was not expected to contribute directly to the subsequent synthesis. The resulting classification was subsequently reviewed.

Three safeguards were applied to maintain classification precision. First, publications were classified according to the Analytics competencies or activities learners were taught, rather than the analytical methods researchers used to evaluate an intervention. Second, technological activities related to constructing, deploying, or maintaining an educational GenAI system were not interpreted as evidence for the corresponding Analytics domain unless learners themselves were being taught that competency. Third, domain assignments required explicit or strongly supported educational evidence rather than assumptions based on broad references to data science, Analytics projects, AI literacy, or GenAI use.

C. Educational Contribution and Practice Synthesis

For each qualifying publication, the analysis identified a source-grounded educational intervention, activity, finding, limitation, or mechanism relevant to the corresponding INFORMS domain. These contributions were synthesized into domain-specific teaching practices describing how GenAI had been used to support learning in that domain. The synthesis was then used to identify recurring practices, instructional patterns, safeguards, and areas in which current educational evidence remains limited.

D. Strategy Extension and Synthesis

The literature synthesis establishes how GenAI is currently applied to Analytics education across the different INFORMS Analytics domains. Based on the identified practices and gaps, complementary strategies are subsequently proposed for each domain to extend the educational uses represented in the corpus. Gaps were identified by comparing the educational contributions observed in the qualifying studies with the competencies represented by the corresponding INFORMS domain. These strategies were formulated to address an identified gap within the corresponding domain, remain aligned with the Analytics competency represented by that domain, and preserve learner responsibility and verification of GenAI-supported work. The proposed strategies are non-exhaustive examples of how the identified gaps can be addressed, not a definitive set of interventions for each

domain. Literature-derived and author-proposed strategies are explicitly distinguished. Finally, both groups are organized as a domain-oriented set of strategies for GenAI-supported Analytics education, connecting Analytics competencies with appropriate GenAI roles and teaching practices while treating learner responsibility, verification, human oversight, and responsible AI use as cross-cutting considerations.

IV. RESULTS

The re-analysis of the 245 publications identified 91 papers containing at least one relevant contribution to GenAI-supported Analytics education according to the classification rubric and evidence thresholds described in Section III. Because classification was multi-label, a publication could contribute to more than one INFORMS domain; therefore, the domain counts are not mutually exclusive. **Table I** summarizes the resulting coverage across the seven domains.

The distribution shows substantial differences in the extent to which current research addresses the competencies associated with each Analytics domain. The largest concentration of contributions occurs in Data and Analytics/Model Development, indicating that GenAI-supported Analytics education has predominantly focused on activities related to data preparation, exploration, analysis, coding, modeling, and evaluation. Business Problem Framing and Methodology Selection receive comparatively less attention, while Analytics Problem Framing and particularly Deployment are sparsely represented. No publication met the established evidence threshold for Analytics Solution Lifecycle Management.

TABLE I
COVERAGE OF GENAI-SUPPORTED ANALYTICS EDUCATION ACROSS THE
INFORMS DOMAINS

INFORMS Domain	Papers
Business Problem Framing	17
Analytics Problem Framing	5
Data	61
Methodology Selection	11
Analytics/Model Development	36
Deployment	3
Analytics Solution Lifecycle Management	0

These results provide the empirical structure for the following domain-oriented strategy development. Rather than treating the uneven distribution solely as a measure of research activity, Section V examines what the identified studies contribute educationally within each domain, synthesizes the teaching strategies supported by the literature, and proposes complementary strategies where current evidence is limited or absent. This domain-by-domain analysis also enables discussion of whether current GenAI integration adequately supports the broader range of competencies required in Analytics education.

V. DOMAIN-ORIENTED TEACHING STRATEGIES

For each INFORMS domain, the analysis first synthesizes teaching practices supported by the qualifying literature and

then identifies limitations or gaps in the available evidence. Complementary teaching strategies are subsequently proposed to address those gaps.

A. Domain 1: Business Problem Framing

Business Problem Framing concerns understanding the organizational context, stakeholders, constraints, risks, and decision that an Analytics initiative is intended to support. Existing studies primarily integrate GenAI through case-based activities that preserve this connection between analysis and organizational purpose. In managerial accounting, students use GenAI while evaluating forecasts and capital-investment decisions under changing market conditions, keeping analytical work tied to a concrete strategic decision [28]. Similar approaches require learners to compare AI-generated forecasting analyses with their own interpretation of market conditions [36], or to evaluate GenAI-supported make-or-buy recommendations while retaining professional judgment over the final business decision [13]. Other studies emphasize stakeholders, constraints, and risks: EthicsAdvisor structures reflection around stakeholders, trade-offs, mitigations, and rationales [37], while revenue-recognition cases require learners to anchor GenAI-supported analysis in contractual terms, governing standards, and the organizational decision about recognizing revenue [38]. Collectively, these practices position GenAI as a means for exploring and questioning business context rather than replacing learner judgment.

However, current strategies largely begin from an already defined case or decision. Less attention is given to teaching students how to determine whether the right business problem has been framed before proceeding to analytical formulation. To address this gap, we propose three complementary strategies. First, a **counterfactual framing exercise** can require GenAI to perturb key contextual conditions, such as budget, regulation, stakeholder priorities, or market conditions, and ask learners to determine whether the original business problem remains valid or must be reframed. Second, a **decision-value traceability activity** can require students to map each proposed business question to a stakeholder action, organizational objective, success criterion, and consequence of an incorrect decision, using GenAI to challenge missing or weak links. Third, a **framing ambiguity audit** can use GenAI to deliberately identify underspecified objectives, hidden assumptions, conflicting stakeholder interests, and unavailable decision information; students must then resolve these deficiencies before any analytical problem is formulated. These strategies extend GenAI from contextual assistance toward explicit development of business-framing competence.

B. Domain 2: Analytics Problem Framing

Analytics Problem Framing translates an organizational need into a precise analytical objective, including the problem type, hypotheses, scope, and criteria by which a solution will be evaluated. Existing evidence provides several useful but relatively narrow instructional practices. In project-based data science, AI-supported templates for classification, regression, and clustering can require learners to identify and justify the analytical problem type before proceeding with

implementation [39]. In statistics education, GenAI has been used to articulate hypotheses and support hypothesis-testing logic while students remain responsible for checking assumptions and alignment with the analytical question [40]. Similarly, macroeconomics students use ChatGPT to formulate and subsequently revise hypotheses according to empirical evidence [41]. Business Analytics optimization instruction provides an especially relevant safeguard by requiring students to formulate the problem independently before comparing it with a GenAI-generated alternative [6]. Prediction exercises further demonstrate the importance of connecting evaluation metrics to the consequences of false-positive and false-negative decisions [42].

Nevertheless, the limited number of studies in this domain indicates that GenAI is more often introduced after an analytical task has already been defined. Less attention is given to teaching students how alternative formulations change the meaning, requirements, and eventual usefulness of an Analytics solution. We therefore propose three complementary strategies. First, an **alternative-formulation challenge** can ask GenAI to generate competing analytical specifications for the same business need, for example, prediction, explanation, segmentation, or optimization, requiring learners to defend which formulation best supports the intended decision. Second, a **formulation stress test** can use GenAI as an adversarial reviewer to expose ambiguous targets, inappropriate units of analysis, unrealistic time horizons, information leakage, or infeasible success thresholds that students must diagnose and correct. Third, a **formulation-to-decision traceability map** can require learners to explicitly connect the analytical objective, target or outcome, scope, evaluation criterion, expected analytical output, and resulting organizational action. Together, these strategies position GenAI not as the source of the analytical formulation, but as a mechanism for comparison, challenge, and refinement while preserving learner ownership of the problem definition.

C. Domain 3: Data

The Data domain contains the largest body of evidence and covers data access, preparation, exploration, quality, visualization, and interpretation. Several studies employ GenAI as a scaffold while preserving independent verification. In civil and environmental engineering education, learners used GenAI while cleaning, visualizing, and analyzing a large air-quality dataset, with structured attention to documenting choices and independently checking generated transformations and interpretations [43]. Data-quality instruction has also combined agentic AI with structured activities for detecting and evaluating quality problems before analysis [44]. Other applications include GenAI-assisted data cleaning and processing with explicit learner responsibility for curation decisions [45], SQL generation accompanied by execution and error correction [10], and personalized synthetic datasets whose distributions and correlations must be checked for instructional consistency [46]. Visualization studies further show the value of active scaffolding: proactive GenAI agents that question learners about visual evidence produced stronger comprehension than passive assistance [47], while adversarial

activities have students create and critique misleading AI-supported visualizations [48].

Despite this breadth, current strategies concentrate mainly on manipulating, exploring, and interpreting data already available to learners. Comparatively less attention is given to teaching whether the available data are appropriate and sufficient for the analytical purpose, how access decisions should be justified, and how limitations introduced upstream constrain subsequent conclusions. We therefore propose three complementary strategies. First, a **data-access negotiation simulation** can use GenAI as a data owner or steward with incomplete, sensitive, or costly sources, requiring learners to justify requested variables and negotiate access under privacy and resource constraints. Second, a **missingness-mechanism challenge** can present alternative GenAI-generated explanations for missing data and require students to determine which mechanisms are plausible and how each would affect permissible treatments and interpretations. Third, a **data-sufficiency gate** can require learners to defend a proceed, recollect, supplement, or reformulate decision before modeling, while GenAI acts as an adversarial reviewer of coverage, representativeness, granularity, and measurement limitations. These strategies extend GenAI use from data processing toward explicit development of data judgment.

D. Domain 4: Methodology Selection

Methodology Selection concerns choosing analytical techniques that are appropriate for the objective, available data, assumptions, and practical constraints. Existing studies primarily position GenAI as a source of candidate methods whose recommendations must be evaluated by learners. In statistics education, students verify whether LLM-selected tests are appropriate before proceeding with analysis [9], while GenAI-supported applications provide model suggestions and theoretical explanations that learners can interrogate before deciding whether the proposed method fits the intended analysis [49]. GenAI can also expose learners to alternative methodological specifications; prompt-driven statistical workflows allow exploration of different likelihoods and effect scales, but require students to track changing assumptions and avoid uncontrolled model proliferation [18]. In machine-learning contexts, learners can compare AI recommendations with competing approaches using empirical benchmark performance [50], while chemical-engineering instruction juxtaposes data-driven and machine-learning methods with first-principles analysis, process simulation, and computational fluid dynamics, enabling learners to compare their assumptions and suitability for a given analytical problem [51]. These practices reinforce that GenAI recommendations should initiate methodological reasoning rather than determine the method automatically.

However, current evidence largely emphasizes selecting among known candidate techniques after the analytical objective and data have been established. Less attention is given to teaching the boundaries of methodological appropriateness and how changing real-world constraints should alter a methodological decision. We therefore propose three complementary strategies. First, a **constraint-inversion**

exercise can have GenAI progressively modify conditions such as interpretability requirements, latency, sample size, computational resources, or regulatory constraints; learners must determine when their original methodology becomes inappropriate and justify a switch. Second, a **methodological abstention challenge** can present cases in which no available method is sufficiently defensible, requiring students to recognize when additional data, reformulation, or further evidence is preferable to forced model selection. Third, a **decision-sensitivity exercise** can require learners to implement plausible competing methodologies and use GenAI to identify where their recommendations diverge, shifting attention from selecting the highest-performing model toward understanding how methodological choice can change the resulting decision. Together, these strategies develop methodology framing as contextual analytical judgment rather than tool recommendation.

E. Domain 5: Analytics/Model Development

Analytics/Model Development is one of the most represented domains and encompasses implementation, experimentation, validation, debugging, and evaluation of analytical models. Current studies frequently position GenAI as a coding and modeling scaffold while requiring learners to retain responsibility for validation. In statistics coursework, GenAI has been used to support code optimization and simulation design through iterative specification, execution, and inspection of generated code [52]. Business Analytics assignments similarly combine GenAI-supported regression, forecasting, and hypothesis testing with manual Excel analysis and written critique [19]. In machine-learning education, GenAI supports construction and refinement of analytical pipelines [22], while business machine-learning scenarios show that AI-generated code can provide useful implementations and improvement suggestions but still require learner and educator refinement for technical correctness and relevance [53]. Other approaches preserve productive struggle through progressive hinting rather than immediate solution disclosure [31] or establish model-building reasoning before introducing GenAI to refine, extend, and test analytical models [54]. Collectively, these practices emphasize execution, debugging, comparison, and verification rather than accepting generated models as completed solutions.

Nevertheless, many applications still center on obtaining or correcting implementations, leaving opportunities to use GenAI more explicitly for learning experimental reasoning and model behavior. We propose three complementary strategies. First, a **model-failure laboratory** can require GenAI to deliberately introduce subtle defects, such as leakage, inappropriate validation, overfitting, or misleading performance optimization, into otherwise plausible pipelines, with learners diagnosing the failure from empirical evidence. Second, a **model-mutation experiment** can have GenAI systematically modify one modeling decision at a time; students must predict the effect before execution and then explain observed changes in performance and generalization. Third, an **evidence-before-revision protocol** can prohibit

immediate acceptance of GenAI optimization suggestions: learners must first formulate a hypothesis about the model weakness, identify diagnostic evidence, and only then request and evaluate an AI-supported revision. These strategies shift GenAI from implementation assistance toward developing disciplined experimentation, diagnostic reasoning, and understanding of why models behave as they do.

F. Domain 6: Deployment

Deployment concerns integrating a validated Analytics solution into the organizational processes, systems, interfaces, and decisions for which it was developed. This domain is sparsely represented in the literature. One project-based study incorporated ChatGPT from idea generation through deployment, requiring student teams to judge the quality, accuracy, and relevance of AI contributions before incorporating them into the final solution [26]. In industrial education, an LLM-supported smart-factory environment connects predictive analytics with a decision dashboard and quality-control workflows, allowing learners to examine how analytical outputs become operational inputs for energy-related decisions [55]. Similarly, cloud-based social analytics instruction connects APIs, storage, Python, natural-language querying, and interactive visualization into an operational workflow, exposing learners to integration and scalability considerations beyond isolated analysis [56]. These examples demonstrate that deployment can be included in Analytics education, but explicit GenAI-supported pedagogical strategies for this domain remain scarce.

The principal gap is therefore not technical implementation alone, but teaching learners to determine whether, where, and under what conditions an analytical solution is ready to enter practice. We propose three complementary strategies. First, a **deployment-readiness challenge** can have GenAI act as an adversarial reviewer of a proposed solution, generating concerns related to latency, usability, privacy, system compatibility, failure handling, and operational constraints that students must resolve before approving release. Second, a **human-AI decision-boundary exercise** can require learners to specify which outputs may trigger automated actions, which require human approval, and which should never be automated, defending these boundaries against GenAI-generated counterarguments. Third, a **deployment rehearsal** can simulate changing operational conditions immediately before release, such as unavailable inputs, interface failures, capacity constraints, or stakeholder resistance, and require students to revise the integration plan, fallback procedure, and communication strategy. These strategies position deployment as an educational competency involving operational judgment, not merely the technical act of placing a model into use.

G. Domain 7: Analytics Solution Lifecycle Management

Analytics Solution Lifecycle Management concerns the continued evaluation and governance of an Analytics solution after deployment, including performance monitoring, data and concept drift, recalibration, retraining, versioning, documentation, and decisions about modification or

retirement. Unlike the preceding domains, no publication in the analyzed corpus met the established evidence threshold for a GenAI-supported educational contribution in this domain. This absence is notable because Analytics solutions do not remain static after implementation: changes in data, operating conditions, user behavior, or organizational objectives can progressively reduce their validity and usefulness. The lack of qualifying studies therefore suggests that current GenAI-supported Analytics education largely ends before learners confront the continuing responsibilities associated with deployed analytical solutions.

Given this gap, we propose three strategies for extending GenAI-supported learning into lifecycle management. First, a **progressive drift simulation** can expose students to a sequence of GenAI-generated operational scenarios in which data distributions, relationships, or external conditions gradually change. Learners must identify the evidence of degradation, distinguish data drift from concept drift or temporary variation, and determine whether continued use, recalibration, or retraining is justified. Second, a **lifecycle decision board** can use GenAI to generate competing recommendations (retain, retrain, replace, or retire a model) while students defend a decision using predefined performance, business, ethical, and operational criteria. This makes lifecycle intervention itself an assessable analytical judgment. Third, a **model-history reconstruction exercise** can provide learners with GenAI-generated monitoring reports, version histories, incidents, stakeholder feedback, and prior modifications from a simulated deployed model. Students must reconstruct why the solution evolved, identify undocumented or unjustified changes, and propose an auditable next action. Together, these strategies extend Analytics education beyond successful model construction and deployment toward sustained responsibility for the validity, traceability, and continued appropriateness of analytical solutions.

VI. SYNTHESIS AND RESEARCH OPPORTUNITIES

A. From Domain-Specific Strategies to Integrated GenAI-Supported Analytics Education

The domain analysis shows that GenAI-supported Analytics education is currently strongest in Data and Analytics/Model Development, while Analytics Problem Framing, Deployment, and especially Analytics Solution Lifecycle Management remain comparatively underdeveloped. This imbalance suggests that educational integration should move beyond the stages where GenAI most easily generates code, queries, visualizations, or models and instead support the complete set of competencies involved in Analytics work. Existing project-oriented approaches already indicate the value of connecting activities rather than teaching AI use as isolated prompts. For example, linked assignments using Python and ChatGPT APIs culminate in a comprehensive analytical project [57], while research-based learning has incorporated GenAI across instrument design, data analysis, and report development [58].

Building on these practices and the strategies proposed in

Section V, GenAI integration should be adapted to the competency represented by each Analytics domain rather than implemented as a uniform form of assistance. Across the proposed strategies, GenAI assumes different roles while learners retain responsibility for analytical judgment, verification, and the decisions produced through AI-supported work.

B. Human Responsibility, Verification, and Assessment Across Analytics Domains

The proposed strategies treat GenAI as variable support rather than a uniform instructional layer. Its role may shift from simulated stakeholder or critic during problem framing to data assistant, methodological challenger, coding collaborator, deployment reviewer, or lifecycle simulator. However, learner responsibility must remain explicit in every domain. Evidence from information-systems instruction shows the value of establishing a manual solution before ChatGPT-supported revision and critical vetting [59], while spreadsheet-modeling students reported learning by identifying and correcting errors in AI-generated attempts [60]. Assessment should consequently focus on inspectable reasoning and decisions rather than only final outputs. Equivalence-aware assessment of machine-learning programs demonstrates how GenAI can evaluate modeling intent despite differing code structures [61], and an Ethics Decision Log illustrates how stakeholder reasoning, evidence, uncertainty, and revisions can become assessable artifacts [62]. Across domains, verification should therefore match the type of AI contribution: contextual claims require evidence, data transformations require inspection, methodological recommendations require justification, models require execution and validation, and operational decisions require human approval and traceability.

C. Educational Implications

For Analytics curricula, the proposed strategies imply that GenAI literacy should be embedded within disciplinary competence rather than taught only as a separate prompting skill. Students should learn not merely to obtain useful outputs, but to determine when AI assistance is appropriate, what must remain under human control, and what evidence is required before an output can inform the next analytical stage. This also changes the instructor role from policing tool access toward designing bounded human-AI activities, checkpoints, and assessment artifacts. Reflective machine-learning laboratories already propose combining AI-assisted coding with unaided work, reflective reports, and assessment checklists to counter an illusion of competence [63]. Similarly, authentic cases co-created with GenAI require instructor review because generated narratives may contain fabricated citations or weak contextual fidelity [64]. These findings reinforce human oversight and responsible AI use as cross-cutting principles across the proposed teaching strategies, together with learner agency and verification.

D. Research Opportunities Toward Evidence-Based Integration

Future research should evaluate the proposed strategies at both domain and end-to-end levels. First, empirical work is particularly needed in Analytics Problem Framing, Deployment, and Analytics Solution Lifecycle Management, where the present corpus provides little or no educational evidence. Second, studies should compare alternative GenAI roles and levels of assistance within the same domain, examining whether strategies that promote critique, adversarial review, or simulation produce stronger transferable competence than direct answer generation. Third, research should assess complete Analytics learning sequences to determine whether competencies developed with GenAI transfer across domain boundaries and persist when assistance is reduced. Finally, evaluation should distinguish artifact quality, learner perceptions, and actual competence, including the ability to work independently, detect AI errors, justify analytical decisions, and manage deployed solutions over time. Such evidence would allow the proposed strategies to evolve from literature-grounded proposals into empirically validated approaches for systematic GenAI-supported Analytics education.

VII. LIMITATIONS OF THE STUDY

This study presents several limitations. First, the analysis is based on a Scopus-indexed corpus covering publications from 2022-2026; consequently, relevant studies indexed in other databases, published outside the selected period, or emerging after data collection may not be represented.

Second, the research is primarily conceptual, documentary, and methodological. Although the classification procedure applies explicit evidence thresholds, domain assignments and the subsequent synthesis depend on the educational contributions reported in the selected publications.

Third, the complementary teaching strategies proposed for the seven Analytics domains are literature-informed conceptual extensions rather than empirically tested interventions. Likewise, the proposed domain-oriented set of teaching strategies has not been experimentally validated. No longitudinal evaluations, long-term follow-ups, or large-scale implementations have been conducted to assess learning effectiveness, transfer of competencies, or sustained impact. These strategies should therefore be interpreted as a structured basis for instructional design and future empirical investigation rather than as empirically validated pedagogical interventions. Finally, the rapid evolution of GenAI technologies may modify their capabilities, limitations, and appropriate educational roles, requiring periodic reassessment of the proposed strategies.

VIII. CONCLUSIONS

This study advances GenAI-supported Analytics education by developing a domain-oriented set of teaching strategies grounded in the seven domains of the INFORMS Analytics Framework. Re-analysis of 245 publications identified 91

studies with relevant contributions, revealing substantial concentration in Data and Analytics/Model Development and considerably less evidence for Analytics Problem Framing, Deployment, and particularly Analytics Solution Lifecycle Management. Building on these findings, the study synthesized literature-derived practices and proposed complementary strategies to address underdeveloped competencies across the complete Analytics process. Together, these contributions support a domain-oriented set of teaching strategies in which GenAI assumes different educational roles according to the Analytics competency being developed, while instructors explicitly define learner responsibility, verification mechanisms, and evidence of learning. The resulting perspective moves GenAI integration beyond task automation toward structured human-AI analytical learning. Across all domains, effective educational integration requires preserving learner agency, analytical judgment, verification, human oversight, and responsible AI use. The proposed strategies therefore provide a structured foundation for designing and subsequently evaluating GenAI-supported Analytics education across the broader lifecycle of Analytics work.

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